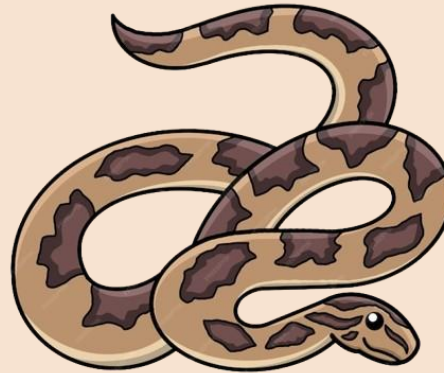
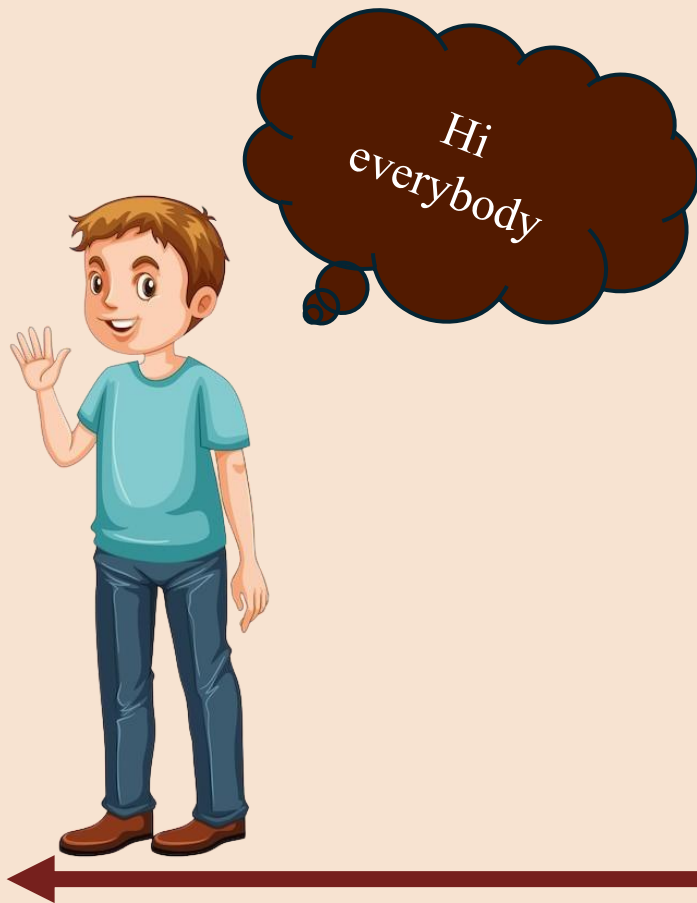
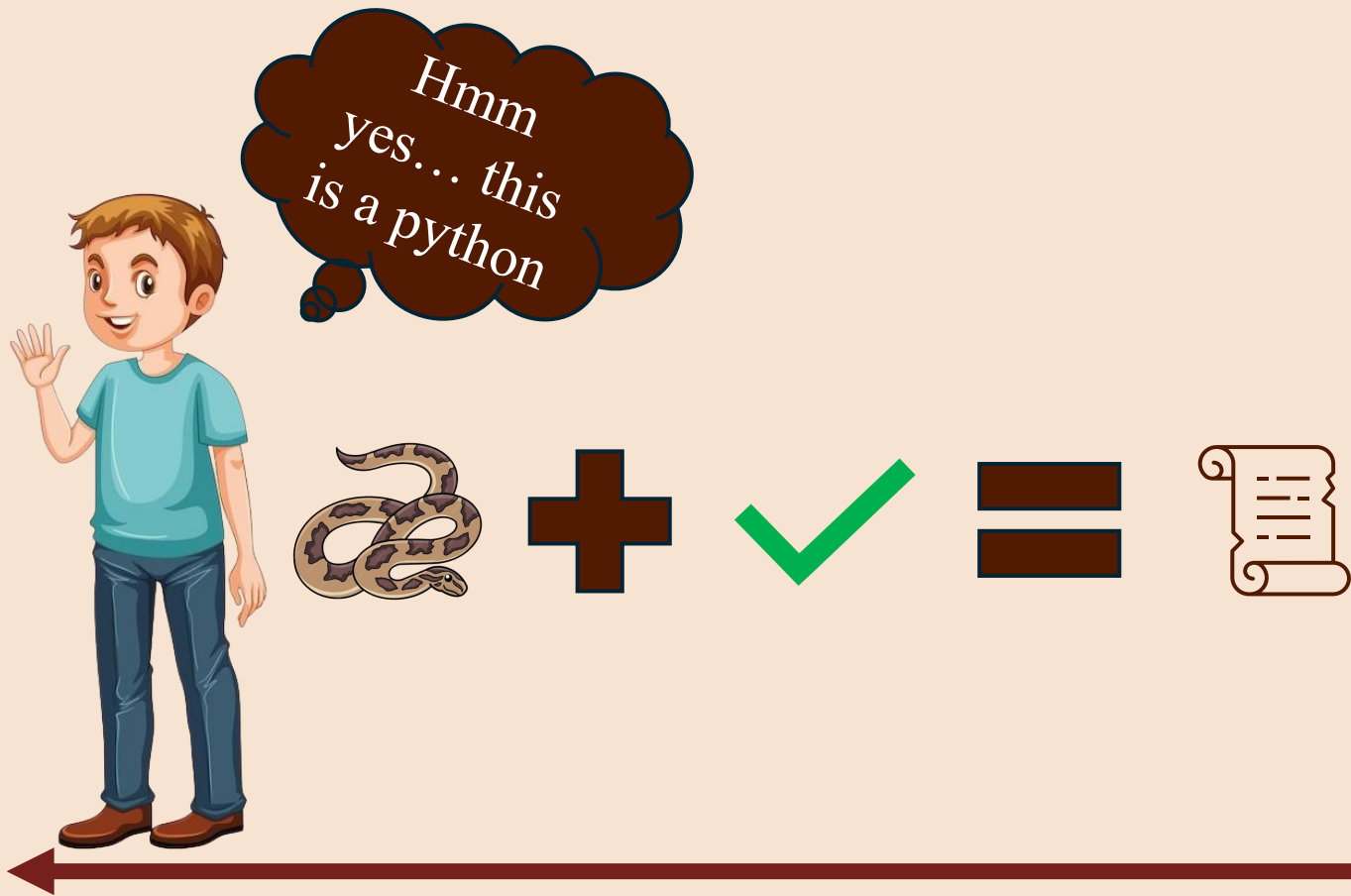


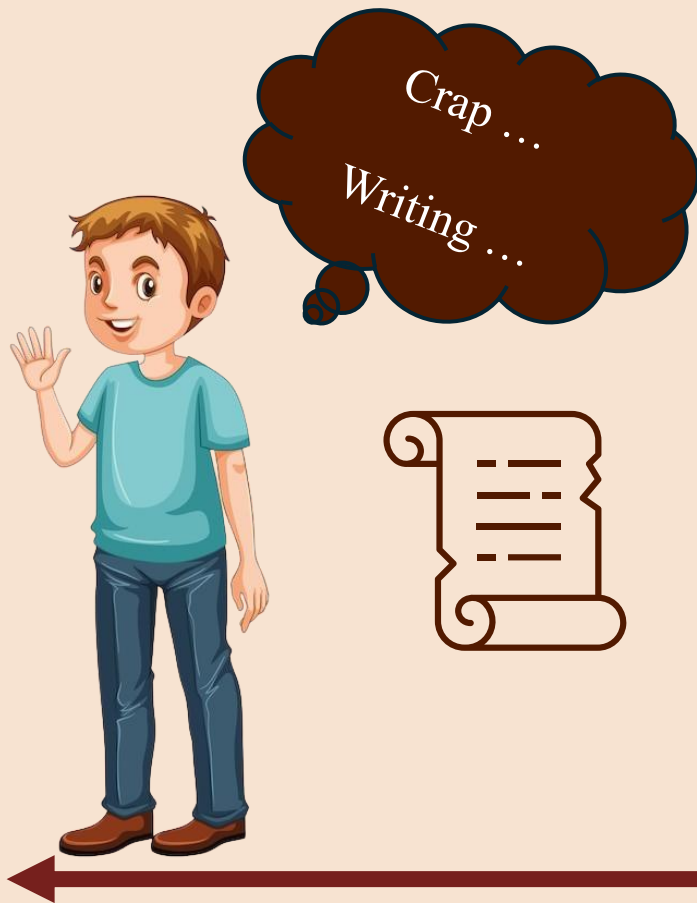
The Role of Temporal Orientation and Subjective Remaining Life in Procrastination

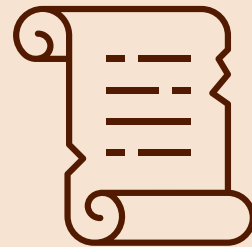
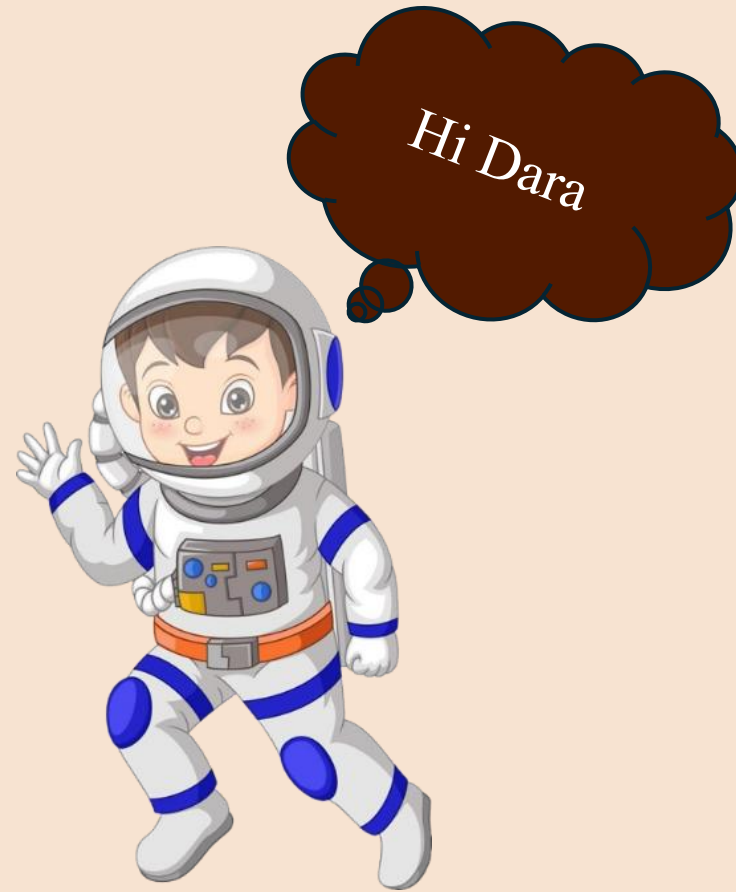
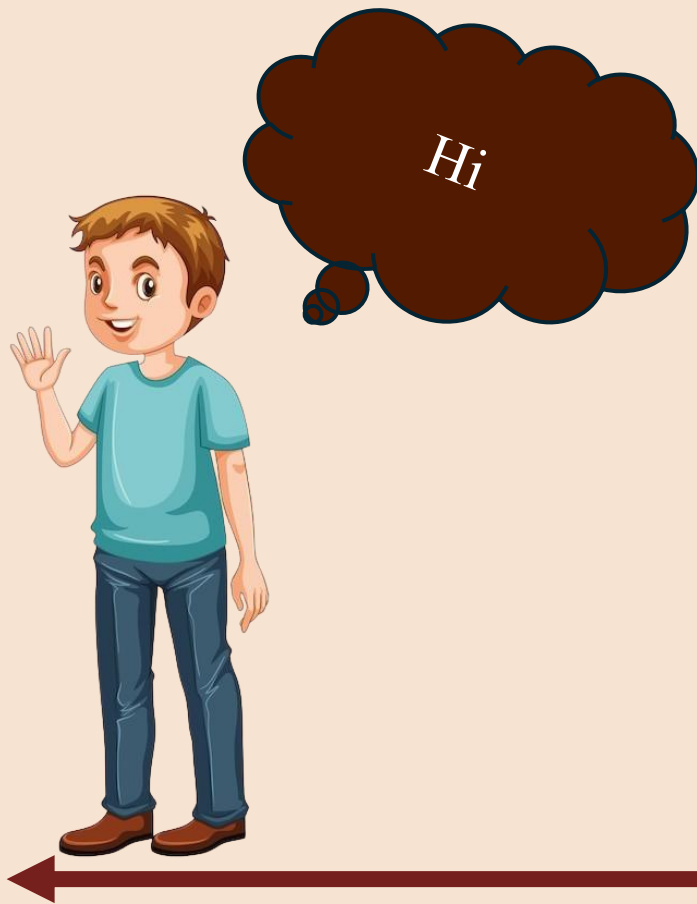
Cormac Monaghan, Ciarán Gartlan, Rafael De Andrade Moral,
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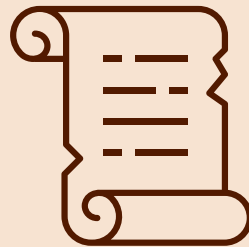


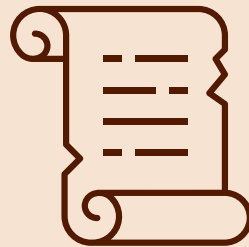


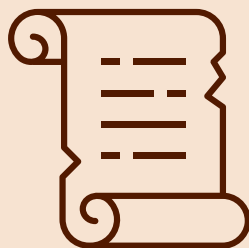


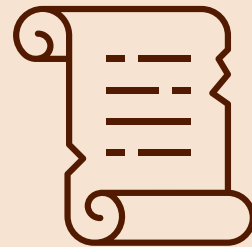


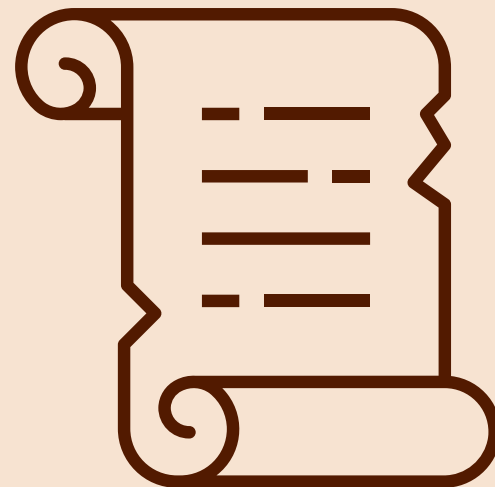
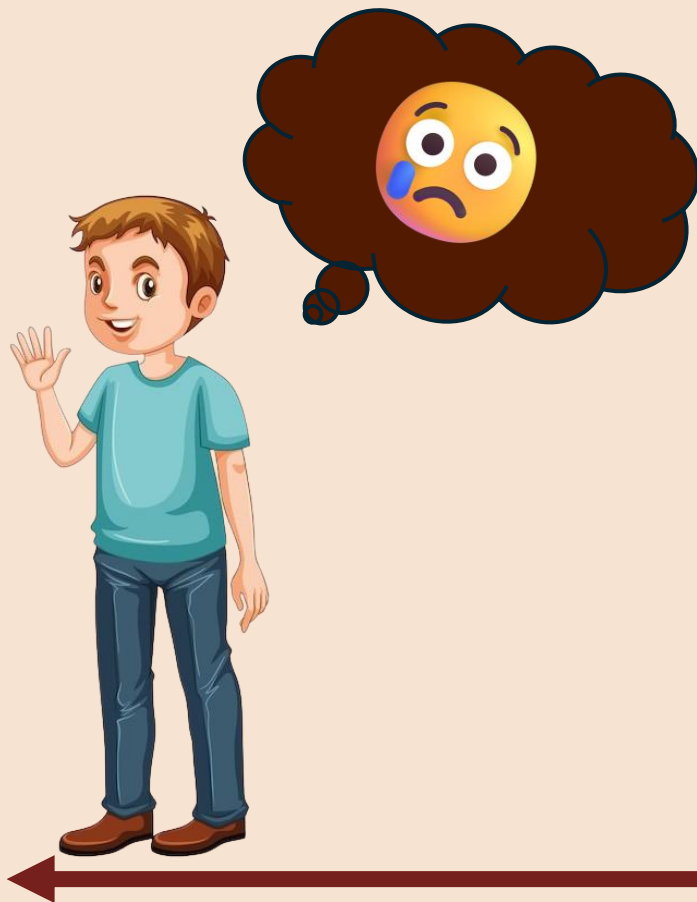
















Self Blame



Guilt

**Low
self-esteem**

Temporal Self-Regulatory Failure

Temporal Orientation

Consideration of Future Consequences



← Immediate
Consequences

→ Future
Consequences

Temporal Orientation

Consideration of Future Consequences

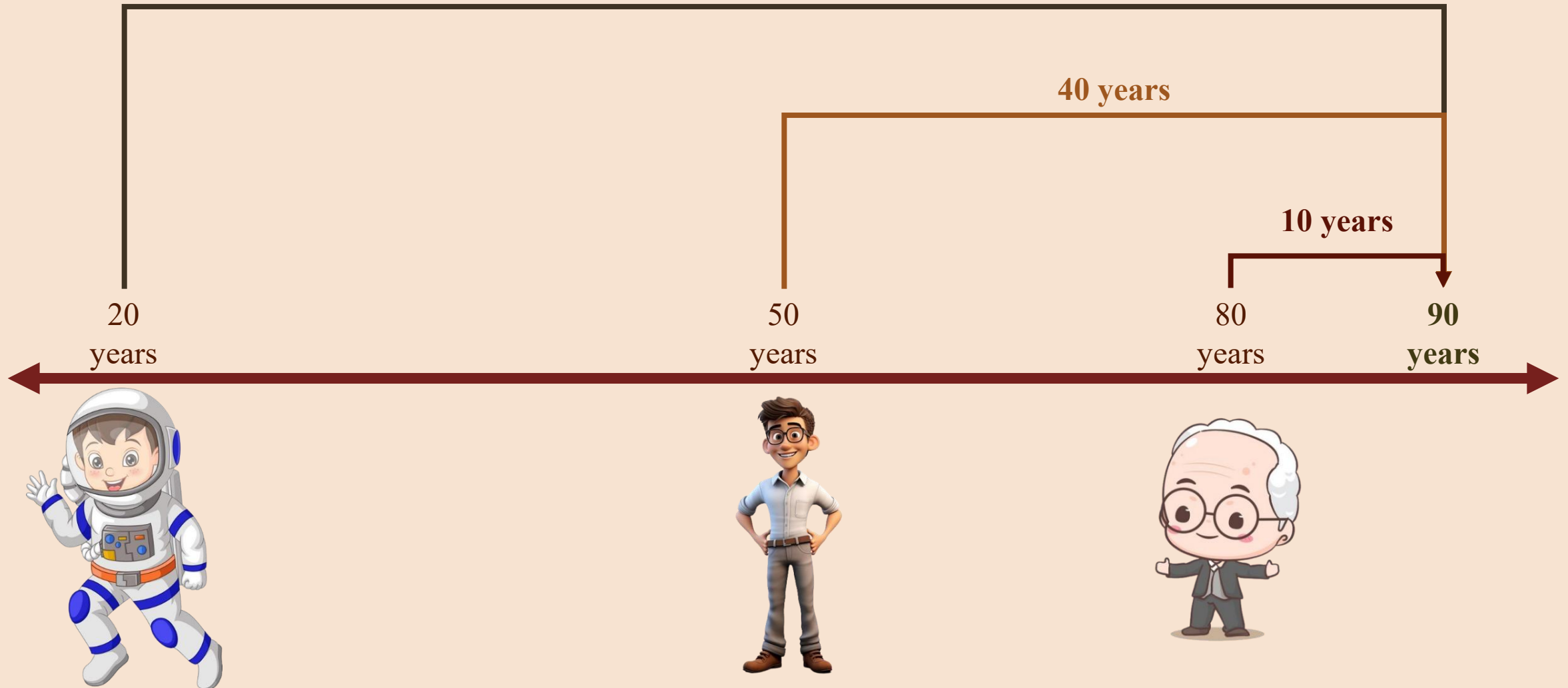


Immediate
Consequences

Future
Consequences

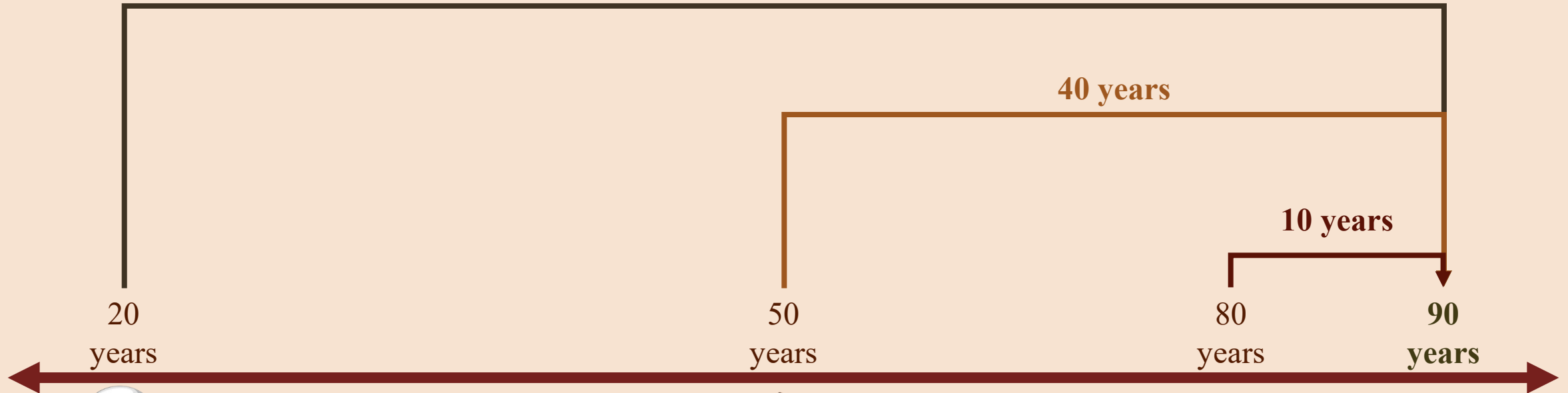
Subjective Remaining Life

70 years



Subjective Remaining Life

70 years



Immediate
Consequences



Does subjective remaining life predict procrastination



Future
Consequences

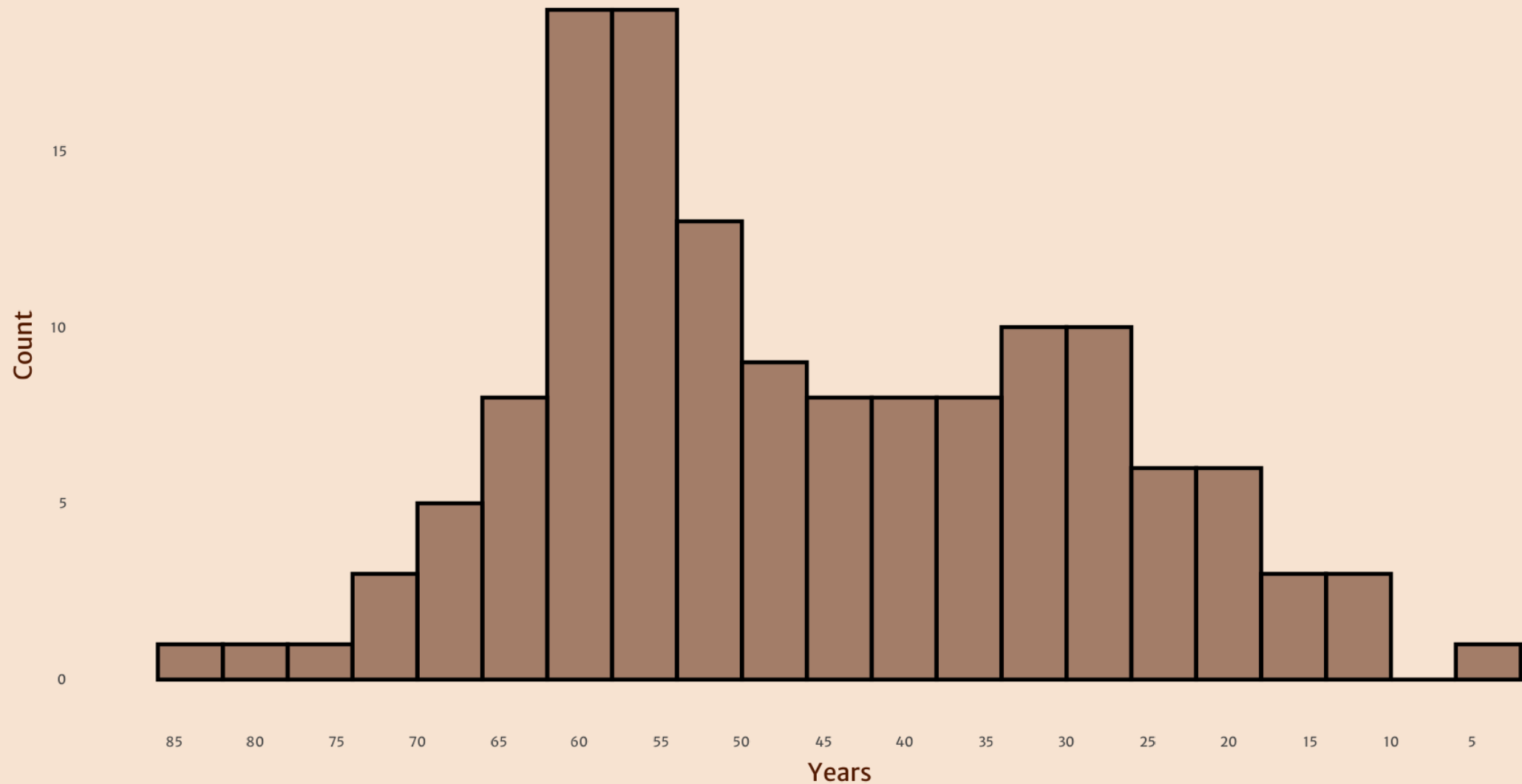
Data Collection

Data was collected by Ciarán

- Temporal Orientation: Consideration of Future Consequences
- Procrastination: Pure Procrastination Scale
- Subjective Life Expectancy
 - Subjective Remaining Life = Subjective Life Expectancy – Age

Subjective Remaining Life

Subjective Life Expectancy - Age



How do we classify someone's temporal orientation



Immediate
Consequences

Future
Consequences

How do we classify someone's temporal orientation



How do we classify someone's temporal orientation



Immediate
Consequences

Future
Consequences

How do we classify someone's temporal orientation



How do we classify someone's temporal orientation



How do we classify someone's temporal orientation



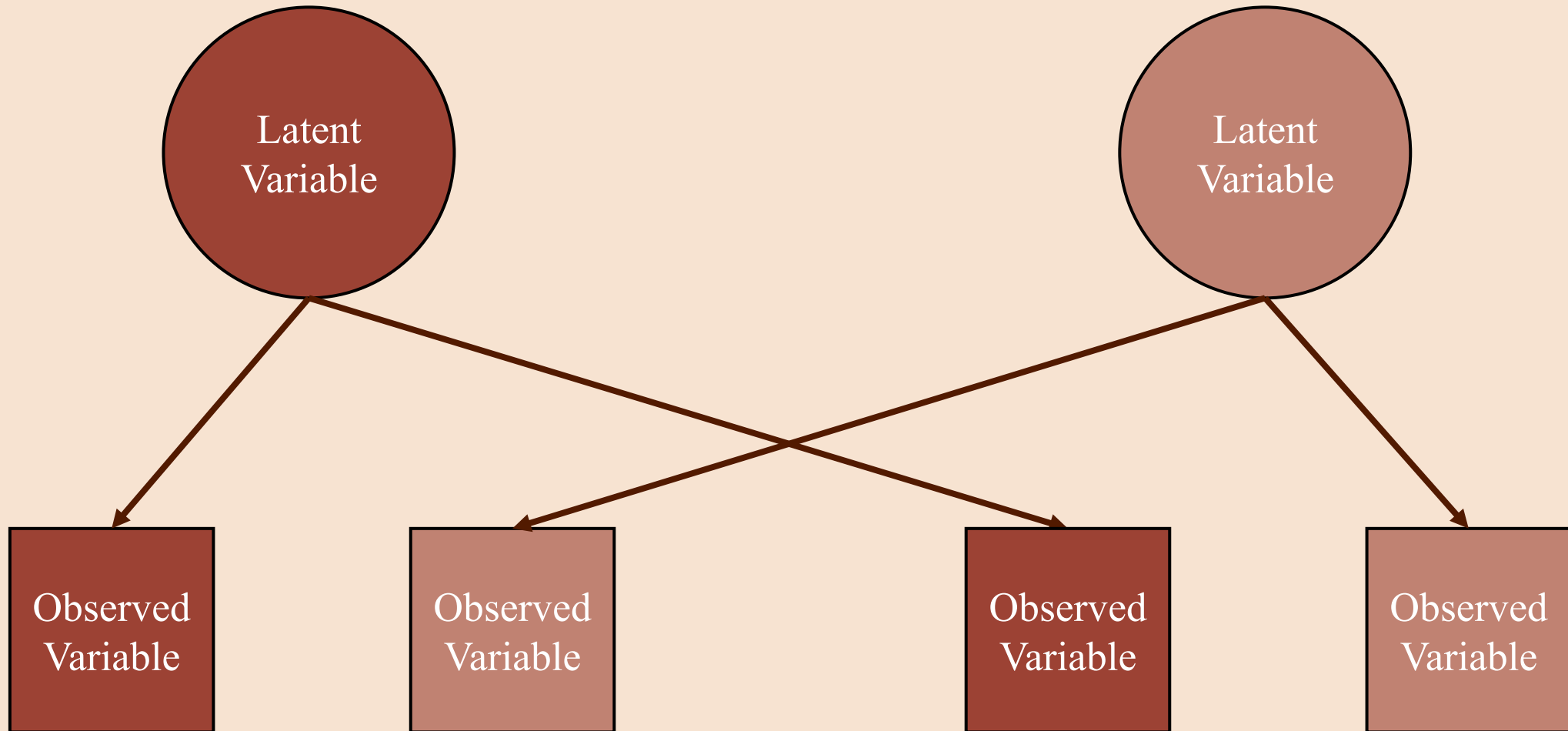
How do we classify someone's temporal orientation



How do we classify someone's temporal orientation



Latent Class Analysis



Latent Class Analysis

LCA assumes that the observed differences in responses are due to the presence of these latent subgroups, otherwise known as **classes** → which we can denote as r

The probability that an individual in latent class r will choose the k -th outcome to the j -th variable is denoted as π_{jrk} .

ID	Item 1	Item 2	Item 3	⋮	Item j
1	1	2	3	⋮	2
2	3	1	3	⋮	1
3	2	1	2	⋮	1
4	2	3	2	⋮	3
5	1	2	2	⋮	2

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$$f(Y_i; \pi_r) = \prod_{j=1}^J \prod_{k=1}^{K_j} (\pi_{jrk})^{Y_{ijk}}$$

$$Y_{ijk} = \begin{cases} 1 & \text{if respondent } i \text{ gives } k\text{-th response to } j\text{-th variable} \\ 0 & \text{otherwise} \end{cases}$$

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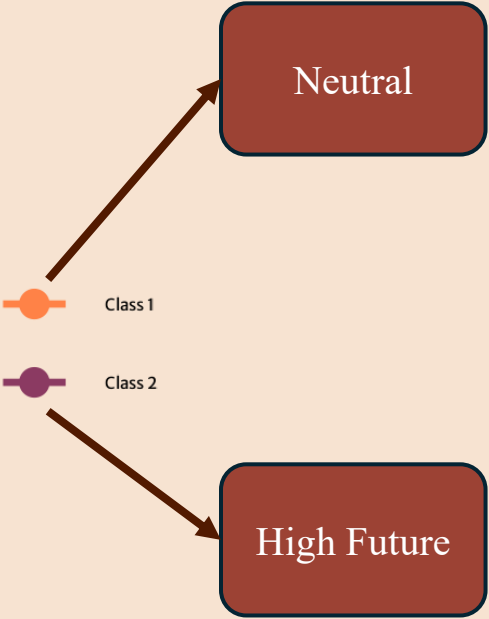
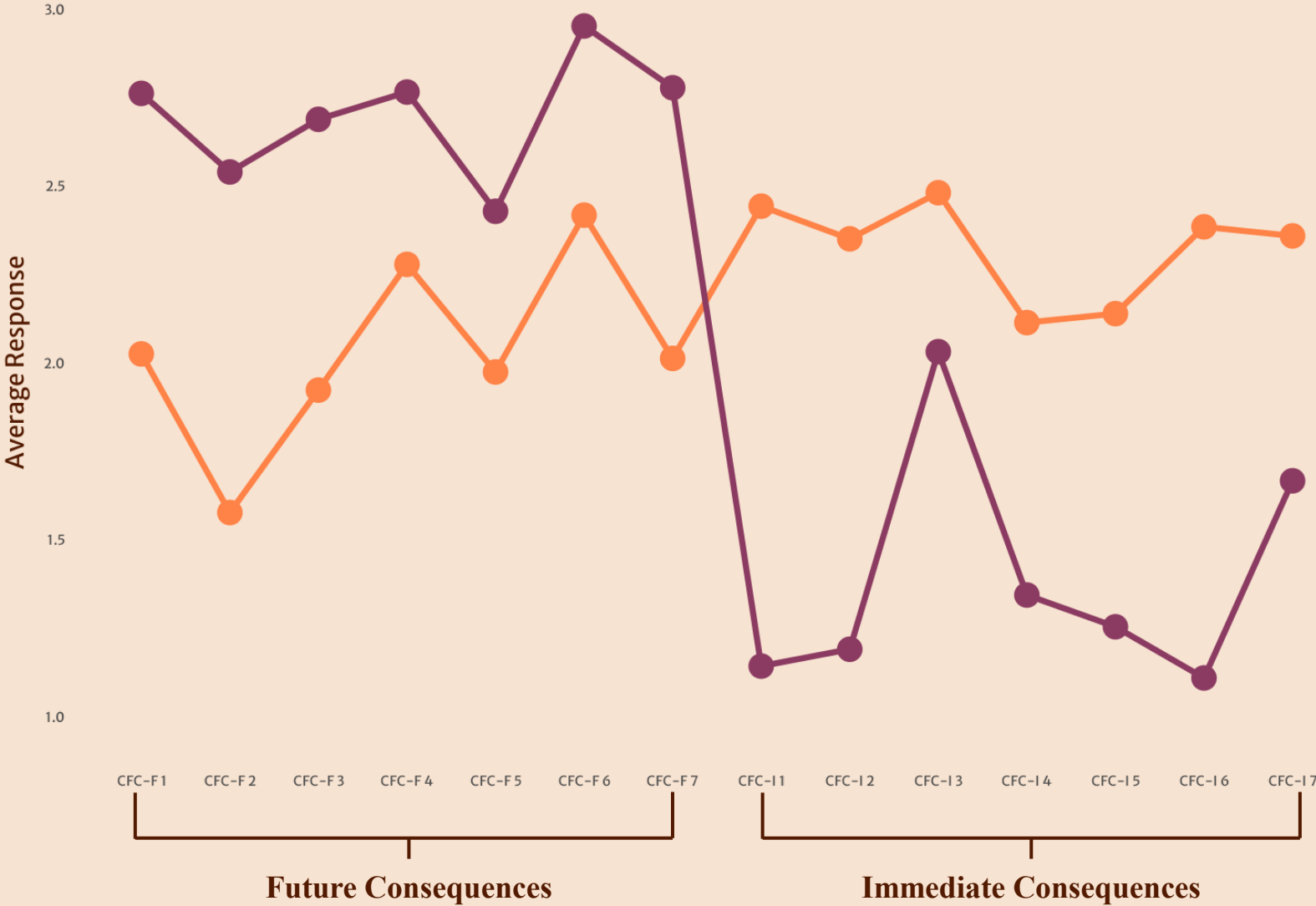
Finally, given estimates $\hat{\pi}_{jrk}$ and $\hat{\gamma}_r$, the posterior probability that an individual belongs to class r , given their observed responses, is calculated using Bayes' theorem:

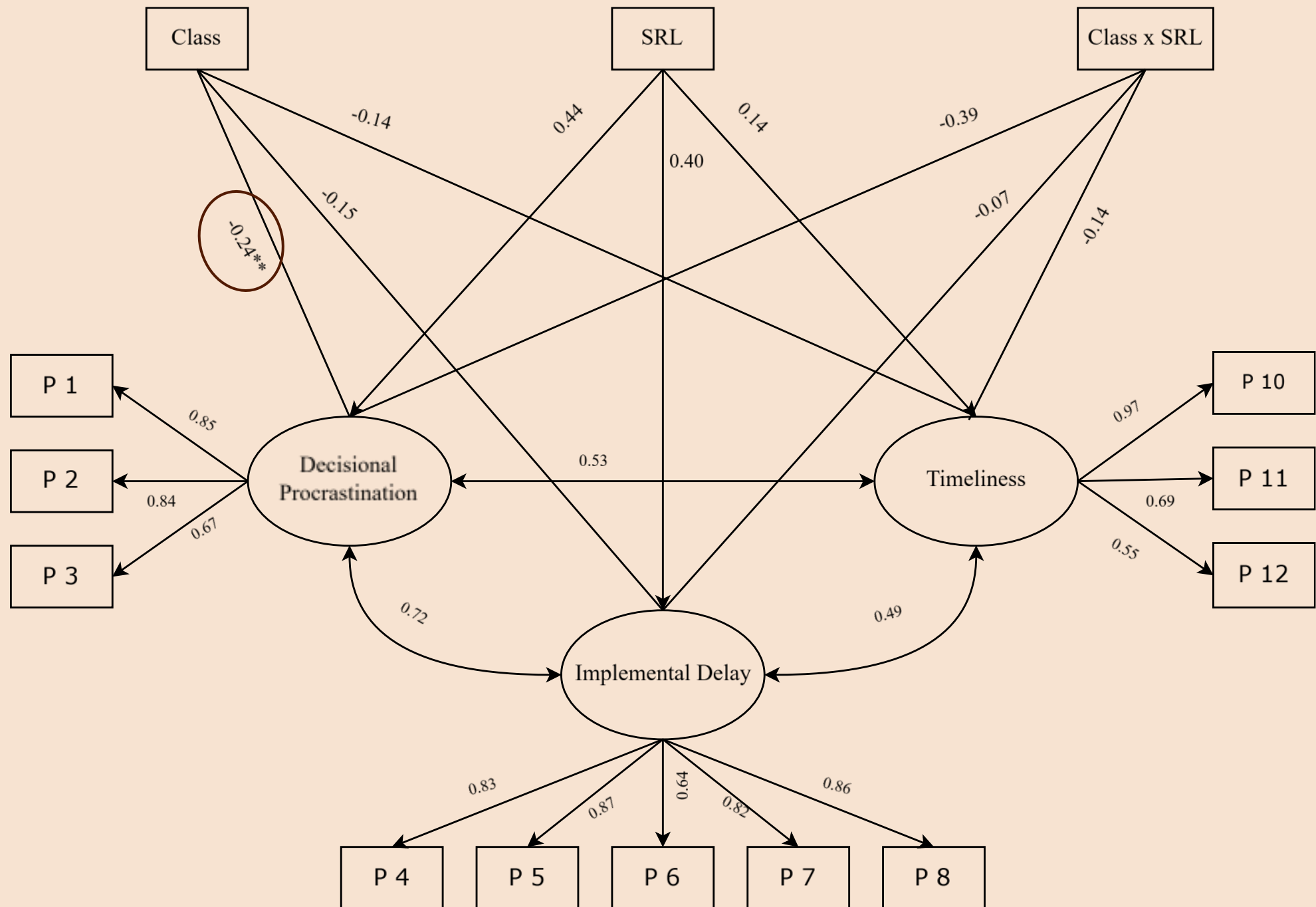
$$\hat{P}(r \mid Y_i) = \frac{\hat{\gamma}_r f(Y_i; \hat{\pi}_r)}{\sum_{q=1}^R \hat{\gamma}_q f(Y_i; \hat{\pi}_q)}$$

ID	Item 1	Item 2	⋮	Item j	P_1	P_2		Classification
1	1	2	⋮	2	0.99	0.01	$\longrightarrow \text{argmax}(p_r) \longrightarrow$	1
2	3	1	⋮	1	0.86	0.14		1
3	2	1	⋮	1	0.20	0.80		2
4	2	3	⋮	3	0.14	0.86		2
5	1	2	⋮	2	0.33	0.67		2

Latent Profile Plot of Item Responses

Class 1 membership = 54.9%, Class 2 membership = 45.1%





Conclusions

- Our findings suggest that a strong future orientation is associated with lower levels of decisional procrastination but has no significant effect on implemental delay or timeliness.
- Subjective perception of time may not be as influential as temporal orientation in shaping procrastination behavior

Limitations

- Cross sectional
- Small sample size (n = 143)

Amazing Paper Title I Will Come Up With Eventually

Cormac Monaghan ^{1,2}, Ciarán Gartlan², Rafael de Andrade Moral ³,
and Joanna McHugh Power ²

¹Hamilton Institute, Maynooth University

²Department of Psychology, Maynooth University

³Department of Mathematics and Statistics, Maynooth University

Temporal Orientation and Procrastination: Exploring the Role of Subjective Remaining Life

AUTHORS

Cormac Monaghan 

Ciarán Gartlan

Rafael de Andrade Moral 

Joanna McHugh Power 

AFFILIATIONS

Hamilton Institute, Maynooth University, Ireland

Department of Psychology, Maynooth University

Department of Psychology, Maynooth University

Department of Mathematics and Statistics, Maynooth University

Department of Psychology, Maynooth University

Abstract

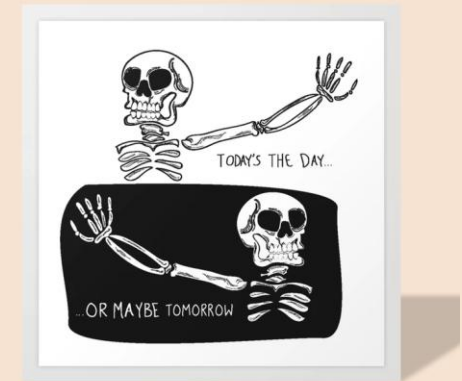
Procrastination is associated with a diminished future time perspective, where individuals prioritize immediate needs over long-term goals. However, individuals who perceive they have less time remaining in life may be less inclined to delay important actions. The purpose of this study was to explore how temporal orientation and subjective remaining life influence procrastination. This secondary analysis uses cross-sectional survey data of 143 participants aged 18-77. Participants completed measures of subjective life expectancy, consideration of future consequences, and procrastination. Latent class analysis and structural equation modelling were applied to the data to examine the potential moderating effects of temporal orientation and subjective remaining life on procrastination. Latent class analysis identified two temporal orientation classes: “neutral orientation (54.9%) and “highly future oriented” (45.1%). These classes were used in a structural equation model to assess moderation by subjective remaining life. The results revealed no moderating or direct effect of subjective remaining life on procrastination. However, high future orientation was negatively associated with decisional procrastination. Subjective perceptions of



Procrastination

Till Death Do Us Part

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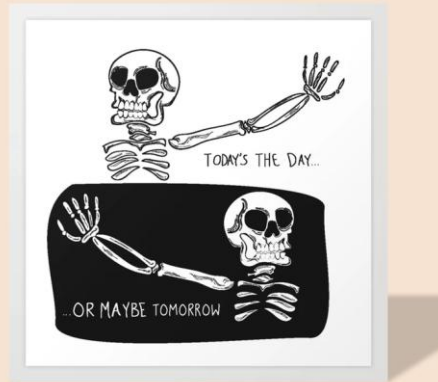




Procrastination

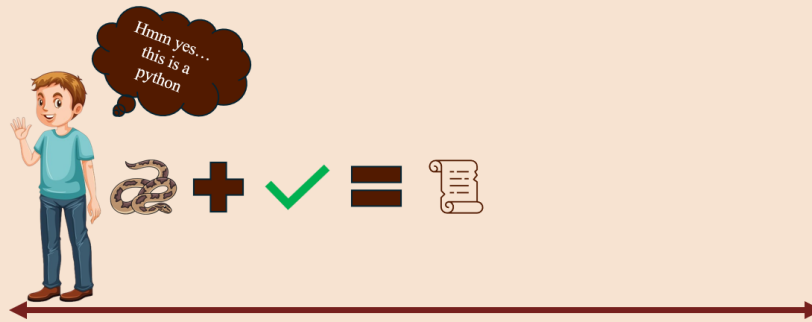
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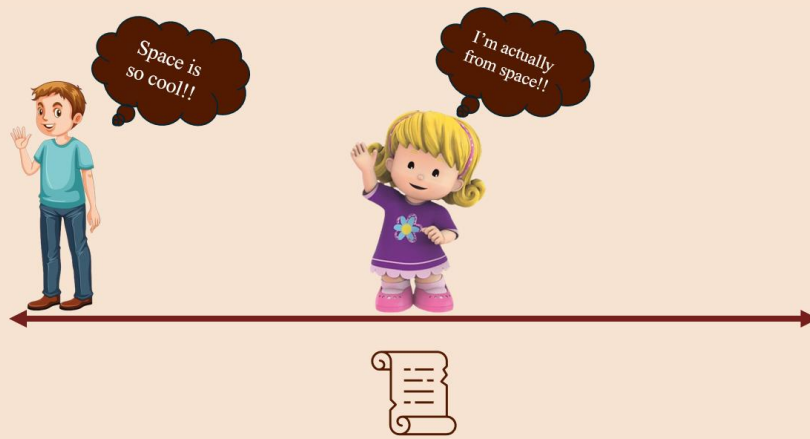




Sirois, F., & Pychyl, T. (2013). Procrastination and the priority of short-term mood regulation: Consequences for future self. *Social and personality psychology compass*, 7(2), 115-127. <https://doi.org/10.1111/spc3.12011>



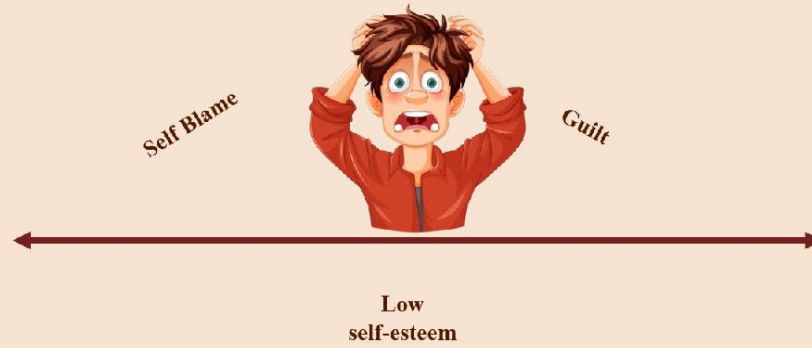
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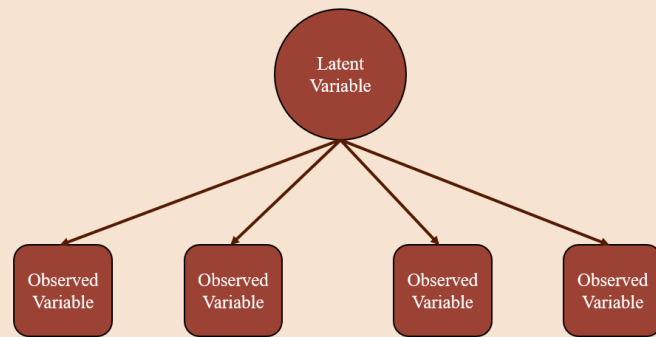
Sirin, F. & Pychyl, T. (2013). Procrastination and the priority of short-term mood regulation: Consequences for future self. *Social and personality psychology compass*, 7(2), 115-127. <https://doi.org/10.1111/spc3.12011>



Baron, R. M., & Preacher, K. J. (2013). Moderator-mediator variable models: Mediation and the power of short-term mood regulation: Consequences for future self. *Social and personality psychology compass*, 7(2), 112-127. <https://doi.org/10.1111/apsd.12011>



Latent Class Analysis





Latent Class Analysis

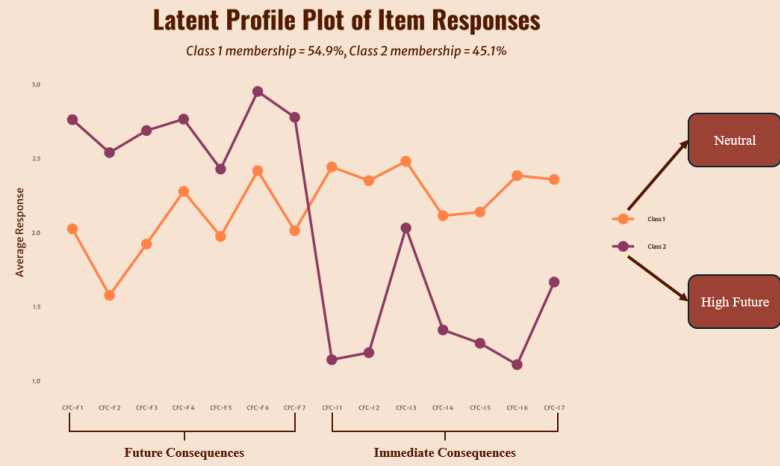
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π_{jrk}
 γ_r → **Expectation Maximization
algorithm**

Aspar, D. A., & Lewis, J. B. (2011). polCA: An R package for polytomous variable latent class analysis. *Journal of statistical software*, 42, 1-29. <https://doi.org/10.18637/jss.v042.i10>





Conclusions

- Our findings suggest that a **strong future orientation** is associated with **lower levels of decisional procrastination** but has no significant effect on implemental delay or timeliness.
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Limitations

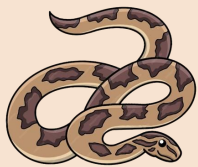
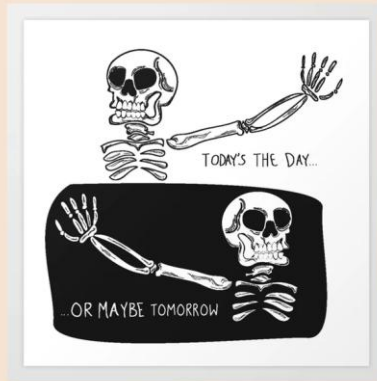
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Procrastination

Till Death Do Us Part

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Questions



Trichotomizing the CFC-Scale

When doing LCA it is important to be aware that the number of estimated parameters increases rapidly with R , J , and K_j .

We can represent our degrees of freedom as:

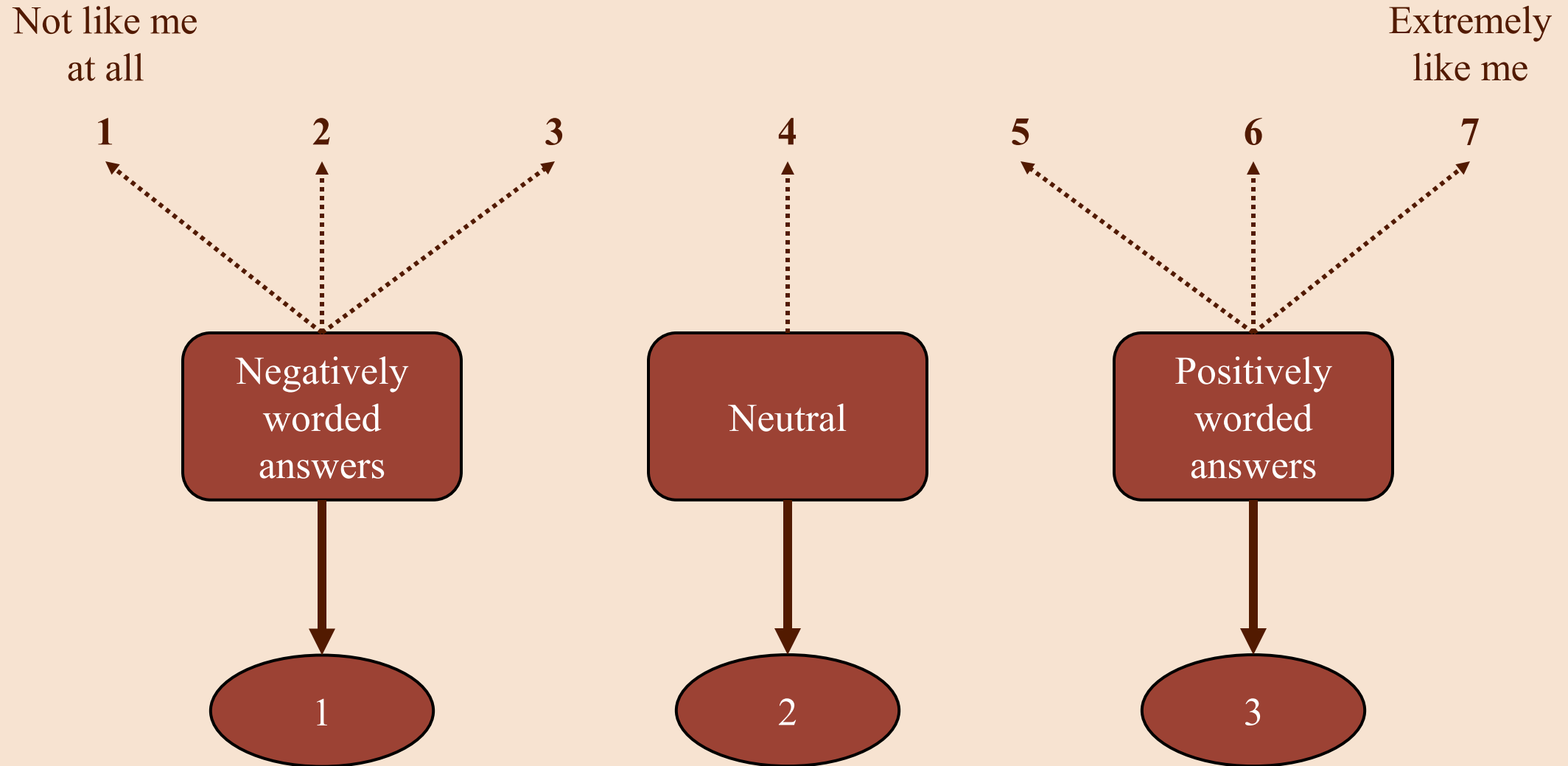
$$DF = N - [R \sum_j (K_j - 1) + (R - 1)]$$

Substituting in values $N = 143$, $J = 14$, and $K_j = 7$ our degrees of freedom are calculated as:

$$DF = 143 - [R \sum_{14} (7 - 1) + (R - 1)] = 144 - 85R$$

Substituting in any value $R > 1$ where $R \in \mathbb{N}$ results in negative degrees of freedom and an unidentifiable model

Trichotomizing the CFC-Scale



EM Algorithm

poLCA estimates the latent class model by maximising the log-likelihood function:

$$\log(L) = \sum_{i=1}^N \ln \sum_{r=1}^R \gamma_r \prod_{j=1}^J \prod_{k=1}^{K_j} (\pi_{jrk})^{y_{ijk}}$$

Arbitrary initial values for $\hat{\pi}_{jrk}$ and $\hat{\gamma}_r$ which are labelled as $\hat{\pi}_{jrk}^{old}$ and $\hat{\gamma}_r^{old}$

$$\hat{\gamma}_r^{new} = \frac{1}{N} \sum_{i=1}^N \hat{P}(r | Y_i)$$

$$\hat{\pi}_{jr}^{new} = \frac{\sum_{i=1}^N Y_{ij} \hat{P}(r | Y_i)}{\sum_{i=1}^N \hat{P}(r | Y_i)}$$

The EM algorithm repeats these steps, updating the parameters until the overall log-likelihood ceases to increment beyond an arbitrary small number

Model selection

Model	K	Log(L)	BIC	Entropy	Proportion of class
1 Class	28	-1967.13	4073.22	NA	-
2 Classes	57	-1771.45	3825.78	0.92	0.549 / 0.451
3 Classes	86	-1708.80	3844.40	0.94	0.428 / 0.155 / 0.417
4 Classes	115	-1669.52	3909.76	0.95	0.303 / 0.419 / 0.144 / 0.164

$$\text{BIC} = -2 \Lambda + \Phi \ln(N)$$

$$\text{Entropy} = 1 - \frac{\sum_{i=1}^N \sum_{r=1}^R [-\hat{P}(r | Y_i) \times \ln(\hat{P}(r | Y_i))]}{N \times \ln(R)}$$